

EXPLORATORY DATA ANALYSIS-EDA

Exploratory → Exploring, Data → Information, Analysis → Understanding the data.

EDA: It is the process of exploring and understanding a dataset before building any ML model. It helps identify patterns, missing values, outliers and relationships in the data.

EDA understands the data, detect missing values, detect outliers, understand distribution, find relationships, generate business insights which will improve the performance of ML model

Types of EDA:

- 1) Univariate Analysis → Study one variable at a time.
- 2) Bivariate Analysis → Study two variable at a time.
- 3) Multivariate Analysis → Study multiple variables at a time.

Data Visualization: It is the process of representing data using graphs and charts so that patterns, trends and relationship become easy to understand.

NOTE: The idea is instead of reading 1000 of lines we can plot it easy to understand them.

MATPLOTLIB: It is a python library used to create static, animated and interactive visualization. In DS about 90% of basic plots are created using matplotlib or libraries built on it.

Line Plot → Use when we want to see change over time.

Bar Plot → Use when we want to compare categories.

Histogram → A histogram is used to understand the distribution of a numerical feature. It helps identify the frequency of values, the shape of the distribution, skewness, outliers, and where most observations are concentrated.

NOTE: If number of bins changes, distribution also changes.

KDE (Kernel Density Estimation) → Is used to estimate and visualize the probability density of numerical feature.

Scatter Plot → Use when we want to find the relationship between two numerical variables.

Pie Chart → Use when we want to see proportion of whole.

Histogram vs Bar Plot ★★★★★	
This is one of the most common interview questions.	
Histogram	Bar Plot
Numerical data	Categorical data
Shows distribution	Shows comparison
Bars touch each other	Bars have gaps
X-axis = intervals (bins)	X-axis = categories
Order matters	Categories can be rearranged

SEABORN: It is a python visualization library built on top of Matplotlib. Seaborn doesn't replace matplotlib, it uses matplotlib internally but provides:

- 1) Better look graphs
- 2) Less Code
- 3) Better support for pandas Data Frames
- 4) Built-in statistical visualizations

Count Plot→ A **Count Plot** is a graph that **counts how many times each category appears** in a dataset.

Count Plot vs Bar Plot	
Count Plot	Bar Plot
Counts categories automatically	You provide the values
Works only with categorical data	Works with categories and numerical values
No aggregation needed	Requires aggregation (count, mean, sum, etc.)
Seaborn	Matplotlib/Seaborn

Box Plot→ A **Box Plot** is a graph that summarizes the distribution of a numerical feature using **five important values**. Minimum, Q1, Median (Q2), Q3, Maximum

Pair Plot→ A **Pair Plot** is a collection of plots that shows the relationship between **every numerical feature** in a dataset.

Heatmap→ Find correlation between features

Violin Plot→ To understand distribution along with spread, **NOTE: Think of it as a BOX PLOT + Smoothed HISTOGRAM.** A VP tells you where the data is and how densely it is packed. **NOTE: VP also tells us where the data is more concentrated.**

4. Box Plot vs Violin Plot

Box Plot	Violin Plot
Shows median	Shows median
Shows quartiles	Shows quartiles
Shows outliers	Shows outliers
Shows spread	Shows spread
✗ Doesn't show distribution shape	✓ Shows distribution shape
Simple	More informative

SUMMARY STATISTICS:

- A) MEASURE OF CENTRAL TENDENCY → These tell you the center of data.
Mean, Median, Mode
- B) MEASURE OF DISPERSION → These tell you how spread out the data is.
Range, Variance, Standard Deviation, Quartiles, IQR
- C) SHAPES OF DISTRIBUTION →
Skewness (Is the data symmetric, is it left skewed, is it right skewed), Kurtosis
(How heavy the tails, Are there many extreme values)

Some Interpretation:

- If $\text{Mean} > \text{Median}$: The data is right-skewed
- If $\text{STD}=2$ Data is highly packed, if $\text{STD}=70$ Data is highly spread

MEAN is sensitive to outliers (Mean depends on the value of every observation) and skewed data, so MEDIAN is often preferred in those cases. MEDIAN depends only on the position of values, not their magnitude, Extreme values does not significantly affect this.

If $\text{MEAN} > \text{MEDIAN} > \text{MODE}$ then it is said to be positive skew (Right Skew)

If $\text{MEAN} < \text{MEDIAN} < \text{MODE}$ then it said to be negative skew (Left Skew)

KURTOSIS measures the heaviness of the tails of the distribution. High kurtosis indicates greater likelihood of extreme values (Outliers)

DATA INSIGHTS: DI are meaningful conclusion drawn from the data analysis that help explain patterns, trends, and relationships, enabling informed and actionable recommendations.

Types Of DATA INSIGHTS:

- 1) Pattern Insights → Look for repeated relationships. Eg: More study hours: Higher Marks.
- 2) Trend Insights → Look for changes over time. Eg: Sales increase every month
- 3) Relationship Insights → Look for connection between variables. Eg: Petal Length increases with petal width.

Data → EDA → Observation → Insight (Why Does It Matter) → Action (What Should We Do) → Decision.

CORRELATION ANALYSIS: It is a statistical technique used to measure the strength and direction of the relationship between two variables.

MULTICOLLINEARITY → Occurs when two or more features are highly correlated, meaning they share provide similar or redundant information.

VIF (Variance Inflation Factor): Is a statistical measure used to detect multicollinearity among independent variables in the dataset.

CALCULATING CORRELATION COEFFICIENTS: It is a numerical value that measures the strength and direction of the relationship between two variables. It ranges from -1 to +1.

TYPES OF CORRELATION COEFFICIENTS:

- A) **PEARSON CORRELATION:** It measures the strength and direction of linear (The point roughly follows a straight line) relationship between two numerical variables.
- B) **SPEARMAN CORRELATION:** It measures the strength and direction of monotonic (A relationship is MONOATONIC if it moves in only one direction) relationship using the rank of data rather than actual values.

Pearson is answering this question:

"How strongly are these two variables linearly related?"

Spearman is answering a slightly different question:

"How strongly do these two variables move together in the same order, even if the relationship isn't perfectly straight?"

NOTE: All linear relationships are monoatonic but not vice versa.

COVARIANCE: It measures whether 2 variables move together. Eg: As experience increases, what happens to salary? It also increases, so covariance tells they move together.

